PRACTICE EXAM 3 OF 4 · EXAM STRATEGY - ALL DOMAINS

Practice exam 3

Twenty new questions, with four from Application Delivery to match its weight across the set.

QUESTIONS 20 TIME About 30 minutes MODULE 34 of 35

admin@sthithapragna.com

> The booking API Deployment is set to 3 replicas. An engineer deletes one of its Pods by hand.

What creates the replacement Pod?

A The kubelet on the same node

B The kube-scheduler

C etcd, from its stored copy

D The ReplicaSet controller

◆ PAUSE AND DECIDE

What creates the replacement Pod?

A The kubelet on the same node

Runs Pods; does not create them.

B The kube-scheduler

Places new Pods; does not create them.

C etcd, from its stored copy

Stores state; creates nothing.

D The ReplicaSet controller

Sees 2 of 3 and creates one more.

THE SIGNAL

Creating Pods to match a count: the ReplicaSet controller.

> Every change at Quayside is built, tested and ready to release.
>
> A manager approves each move into production.

Which practice is this?

A Continuous deployment

B Continuous delivery

C Continuous integration only

D Infrastructure as code

◆ PAUSE AND DECIDE

Which practice is this?

A Continuous deployment
No human gate at all.

B Continuous delivery
Always releasable, with a human gate.

C Continuous integration only
Stops before any environment.

D Infrastructure as code
About infrastructure, not release.

THE SIGNAL

Human approval before production: continuous delivery.

> Quayside wants each Deployment to record the full URL of the CI build that produced it. Nothing will ever select objects by this value.

Where should it be stored?

A In a label

B In a ConfigMap

C In an annotation

D In the namespace's name

◆ PAUSE AND DECIDE

Where should it be stored?

A In a label

For selection, and limited to 63 characters.

B In a ConfigMap

Feeds apps; does not describe the object.

C In an annotation

Descriptive metadata; long values allowed.

D In the namespace's name

Namespaces organise; they do not record builds.

THE SIGNAL

Describe, never select: annotation.

> The three bookings database replicas must discover each other's individual Pod addresses through DNS.

What should their Service set?

A clusterIP: None, a headless Service

B type: NodePort, one port per node

C type: ExternalName, a DNS alias

D type: LoadBalancer, one cloud IP

◆ PAUSE AND DECIDE

What should their Service set?

A clusterIP: None, a headless Service
DNS then returns each Pod's own IP.

B type: NodePort, one port per node
Opens node ports; still one virtual IP.

C type: ExternalName, a DNS alias
An alias pointing outside.

D type: LoadBalancer, one cloud IP
External, still one balanced address.

THE SIGNAL

Each Pod's own address through DNS: headless.

> Before kernel patching, every Pod on node-7 must move elsewhere.
>
> The booking API's disruption budget must be respected.

Which command does this?

A kubectl cordon node-7

B kubectl drain node-7 --ignore-daemonsets

C kubectl delete node node-7

D kubectl taint nodes node-7 x=y:PreferNoSchedule

◆ PAUSE AND DECIDE

Which command does this?

A kubectl cordon node-7

Stops new Pods; existing ones stay.

B kubectl drain node-7 --ignore-daemonsets

Cordons, then evicts through the Eviction API.

C kubectl delete node node-7

Removes the node object; no graceful eviction.

D kubectl taint nodes node-7 x=y:PreferNoSchedule

A soft preference; evicts nothing.

THE SIGNAL

Move the Pods out, respecting budgets: drain.

> Upgrade 5 of the booking Helm release broke checkout. Revision 3 was the last good one.

Which command returns the release to revision 3?

A helm upgrade booking --revision 3

B kubectl rollout undo release/booking

C helm rollback booking 3

D helm history booking --restore 3

◆ PAUSE AND DECIDE

Which command returns the release to revision 3?

A helm upgrade booking --revision 3
No such flag.

B kubectl rollout undo release/booking
Helm releases are not kubectl rollouts.

C helm rollback booking 3
Rolls back to revision 3, as a new revision.

D helm history booking --restore 3
history lists; it restores nothing.

THE SIGNAL

Helm rollback: helm rollback RELEASE REVISION.

> In the payments namespace, containers that set no resource requests should automatically get sensible default requests.

Which object provides that?

A A ResourceQuota

B A PodDisruptionBudget

C A PriorityClass

D A LimitRange

◆ PAUSE AND DECIDE

Which object provides that?

A A ResourceQuota
Caps totals; sets no defaults.

B A PodDisruptionBudget
Governs voluntary evictions.

C A PriorityClass
Sets scheduling priority.

D A LimitRange
Fills in default requests and limits.

THE SIGNAL

Per-container defaults: LimitRange.

> A developer's kubectl command returns 403 Forbidden. Their identity-provider login worked normally.

Which stage rejected the request?

A Authorisation: known user, no permission

B Authentication: identity not recognised

C An admission webhook that is down

D etcd being unavailable

◆ PAUSE AND DECIDE

Which stage rejected the request?

A Authorisation: known user, no permission
403 means authenticated but not allowed.

B Authentication: identity not recognised
That returns 401.

C An admission webhook that is down
A down webhook gives a different error.

D etcd being unavailable
Storage errors are not 403.

THE SIGNAL

403: I know you, and no.

> A Pod with restartPolicy: Never runs a report script. The container exits with code 0.

What is the Pod's phase afterwards?

A Running

B Failed

C Succeeded

D Pending

◆ PAUSE AND DECIDE

What is the Pod's phase afterwards?

A Running

Nothing is running any more.

B Failed

Exit code 0 is success.

C Succeeded

All containers exited successfully.

D Pending

It already ran.

THE SIGNAL

All containers exited 0, no restart: Succeeded.

> Quayside wants HTTP services that scale to zero when idle, with revisions and traffic splitting between them, on Kubernetes.

Which CNCF project provides that?

A Knative

B KEDA

C Karpenter

D Helm

◆ PAUSE AND DECIDE

Which CNCF project provides that?

A Knative
Serverless Serving: scale to zero, revisions, splits.

B KEDA
Event-driven scaling; no revisions or HTTP routing.

C Karpenter
Adds nodes, not serverless apps.

D Helm
Packages apps; does not scale them.

THE SIGNAL

Serverless HTTP, revisions, scale to zero: Knative.

> An engineer wants the built-in documentation for the strategy field of a Deployment's spec.

Which command shows it?

A kubectl describe deployment --field strategy

B kubectl get deployment --docs

C kubectl help deployment.strategy

D kubectl explain deployment.spec.strategy

◆ PAUSE AND DECIDE

Which command shows it?

A kubectl describe deployment --field strategy
Describes objects, not field docs.

B kubectl get deployment --docs
No such flag.

C kubectl help deployment.strategy
Not how kubectl help works.

D **kubectl explain deployment.spec.strategy**
Documents any field, from the API schema.

THE SIGNAL

Field documentation: kubectl explain.

> Quayside must block privileged containers and host namespaces in app namespaces, while most apps keep working unchanged.

Which Pod Security Standard should be enforced?

A Privileged

B Baseline

C Restricted

D Default

◆ PAUSE AND DECIDE

Which Pod Security Standard should be enforced?

A Privileged
Allows everything.

B Baseline
Blocks known escalations; most apps pass.

C Restricted
Also requires non-root and more.

D Default
Not a Pod Security Standard.

THE SIGNAL

Block the dangerous, keep apps working: Baseline.

> Quayside wants exactly 5% of requests to reach a canary.
>
> The share must not depend on how many Pods each version runs.

What achieves that?

A Weighted backendRefs in an HTTPRoute

B One canary Pod among twenty stable Pods

C Two Deployments, switched with Recreate

D A PodDisruptionBudget on the canary

◆ PAUSE AND DECIDE

What achieves that?

A Weighted backendRefs in an HTTPRoute
Exact weights, independent of Pod count.

B One canary Pod among twenty stable Pods
About 5%, but tied to Pod count.

C Two Deployments, switched with Recreate
Recreate swaps everything at once.

D A PodDisruptionBudget on the canary
Governs evictions, not traffic.

THE SIGNAL

Exact share, independent of Pods: traffic weights.

> On a kubeadm control plane node, the API server runs as a Pod.
>
> It must start before any API server exists to accept Pod objects.

How does the kubelet run it?

A As a DaemonSet in kube-system

B As a Deployment in kube-system

C Only as a systemd service, never a Pod

D As a static Pod, from a manifest on disk

◆ PAUSE AND DECIDE

How does the kubelet run it?

A As a DaemonSet in kube-system
DaemonSets need a running API server.

B As a Deployment in kube-system
Deployments need a running API server.

C Only as a systemd service, never a Pod
kubeadm runs it as a Pod.

D As a static Pod, from a manifest on disk
The kubelet reads the manifest directly.

THE SIGNAL

Control plane Pods before the API exists: static Pods.

> The bookings database's data must survive even if its PersistentVolumeClaim is deleted by mistake.

Which reclaim policy should its StorageClass use?

A Delete

B Retain

C Recycle

D Immediate

◆ PAUSE AND DECIDE

Which reclaim policy should its StorageClass use?

A Delete

Deletes the disk with the claim.

B Retain

Keeps the volume and its data.

C Recycle

Deprecated.

D Immediate

A binding mode, not a reclaim policy.

THE SIGNAL

Data outlives the claim: Retain.

> The control plane runs Kubernetes 1.37. The team checks which kubelet versions are supported with it.

Which kubelet version is supported?

A 1.38

B 1.33

C 1.35

D 1.32

◆ PAUSE AND DECIDE

Which kubelet version is supported?

A 1.38

Newer than the API server: never allowed.

B 1.33

Four minors older: too old.

C 1.35

Two minors older: allowed.

D 1.32

Five minors older: too old.

THE SIGNAL

Kubelets: up to three minors older, never newer.

> Quayside's traces are produced with OpenTelemetry. The team needs a CNCF backend to store and search them.

Which project fits?

A Jaeger

B Prometheus

C Fluentd

D Grafana Loki

◆ PAUSE AND DECIDE

Which project fits?

A **Jaeger**
A graduated tracing backend.

B **Prometheus**
Stores metrics, not traces.

C **Fluentd**
Collects and routes logs.

D **Grafana Loki**
A log store, from Grafana Labs.

THE SIGNAL

Store and search traces: Jaeger.

> A new fare feature is deployed with the code switched off.
>
> Product managers want to turn it on for staff first, then everyone, without redeploying.

What makes that possible?

A A canary release

B A blue-green switch

C Feature flags

D The Recreate strategy

◆ PAUSE AND DECIDE

What makes that possible?

A A canary release
Splits by version, and needs deployments.

B A blue-green switch
Switches whole versions.

C Feature flags
Turn features on per audience, no redeploy.

D The Recreate strategy
Redeploys everything.

THE SIGNAL

Release without redeploying: feature flags.

> The timetable renderer never calls the Kubernetes API. Security wants no API token inside its Pods at all.

What should the team set?

A Delete the namespace's default ServiceAccount

B automountServiceAccountToken: false

C A NetworkPolicy blocking the API server

D An RBAC rule denying its ServiceAccount

◆ PAUSE AND DECIDE

What should the team set?

A Delete the namespace's default ServiceAccount
It is recreated automatically.

B automountServiceAccountToken: false
No token is mounted into the Pods.

C A NetworkPolicy blocking the API server
Restricts traffic; the token is still there.

D An RBAC rule denying its ServiceAccount
RBAC has no deny rules.

THE SIGNAL

No token at all: automountServiceAccountToken: false.

> Every container in the payment worker sets CPU and memory requests exactly equal to its limits.

Which QoS class does the Pod get?

A BestEffort

B Burstable

C Critical

D Guaranteed

◆ PAUSE AND DECIDE

Which QoS class does the Pod get?

A BestEffort
That needs no requests or limits at all.

B Burstable
Some requests or limits, but not equal.

C Critical
Not a QoS class.

D **Guaranteed**
Requests equal limits, for CPU and memory.

THE SIGNAL

Requests equal limits everywhere: Guaranteed.

Score yourself by domain

DOMAIN	QUESTIONS	YOUR SCORE
Kubernetes Fundamentals (44%)	1, 3, 5, 7, 9, 11, 14, 16, 20	of 9
Container Orchestration (28%)	4, 8, 12, 15, 19	of 5
Application Delivery (16%)	2, 6, 13, 18	of 4
Cloud Native Architecture (12%)	10, 17	of 2

ON THE EXAM

Three papers in: the same weak domain each time means rewatch that domain's modules now.

The final paper: practice exam 4, then a trend review across all four.

admin@sthithapragna.com